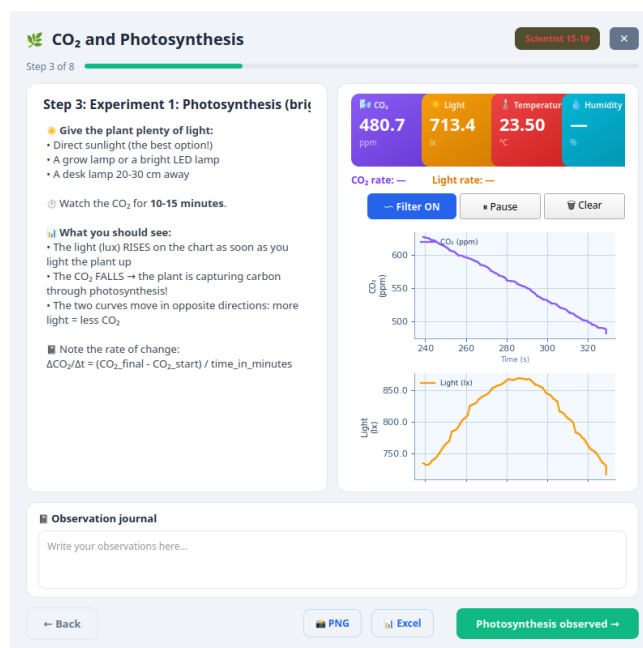


CO₂ and Photosynthesis

A plant in a closed jar — and the point where photosynthesis and respiration cancel exactly

Level: Scientist — ages 15–19 (Romanian school years IX–XII) • **Time in class:** 50 min, or 2–3 lessons • **Sensors:** SCD41 (carbon dioxide) and the light sensor (illuminance)



The activity screen. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Scientist — ages 15–19 (Romanian school years IX–XII). The most direct entry is in **year IX** (biology: photosynthesis and cellular respiration), and the activity gains a second life in **years XI–XII**, where *limiting factors* and the *light compensation point* are explicit syllabus content. In year VIII it can be run as a demonstration, alongside the carbon cycle.

Assumed knowledge

The equations of photosynthesis and cellular respiration. Reading a gradient off a graph and computing a rate, in ppm per minute. For the quantitative analysis: an *XY* chart and the idea of a *plateau*, that is, of a factor that limits.

Time in class

Short: 50 min — theory, rig, photosynthesis in light and respiration in darkness. Full: 2–3 lessons, adding the light-intensity series, the compensation point and the limiting factors.

Teacher preparation

30 min — **choose and try the plant**, because everything depends on it; assemble the jar and check that it seals; **do a ten-minute trial run** to see whether the CO₂ really does fall in light; and prepare the light source and the shading sheets.

Sensor settling time

Ten minutes after closing the jar, before you start recording: the air inside has to mix, and the SCD41 needs thirty seconds merely to start. **Every change of light needs another five minutes** before the gradient settles — the gradient, not the value.

Sensors and boxes	SCD41 for CO ₂ and the light sensor for illuminance; the model in the box is the VEML7700. Both are needed: without measured light the intensity series has no axis. One box per group; two allow a plant-free control , which is the most valuable thing you can add.
Three routes	
Short	50 min: theory (10), the rig and its ten minutes of settling (10), photosynthesis in bright light (15), respiration in darkness (10), the conclusion (5).
Full	2–3 lessons: lesson 1 as above; lesson 2 the light-intensity series and the compensation point ; lesson 3 the limiting factors, the rate calculations and the link to climate change.
Project	Extension: the same rig with a second, <i>plant-free</i> jar as a control, run simultaneously — without it, any rise in CO ₂ can be blamed on a leak.

1. Where this fits in the curriculum

- **Biology** — in **year IX**, plant physiology: photosynthesis and cellular respiration. The syllabus asks for both processes, and here they are seen *competing*, on the same graph, in the same jar. **That a plant respire by day as well as by night is the commonest misconception at this level**, and the activity demolishes it in one lesson.
- **Biology, years XI–XII** — *limiting factors* and the *light compensation point* are measurable quantities here, not definitions: pupils find for themselves the intensity at which the CO₂ line goes flat.
- **Chemistry** — in **year X**, glucose and its reactions; the activity's two equations are exactly the reverse of each other, which shows very well written one under the other on the board.
- **Ecology and environmental science** — in **year VIII**, the carbon cycle; in **years XI–XII**, the role of vegetation in the global carbon budget and the limits of the analogy between a jar and a forest.
- **Mathematics** — the gradient as a rate, in ppm per minute, and a graph that flattens: a first encounter with the idea of *saturation*.

The core skill is **reading a net rate**. The CO₂ in the jar does not measure photosynthesis; it measures the *difference* between photosynthesis and respiration. The pupil who understands that a flat line does not mean “nothing is happening” but “the two processes cancel” has understood the lesson.

2. Time plan

Route	Time	What happens
Short	50 min	Theory and the two equations (10), the rig and its ten minutes of settling (10), photosynthesis in bright light (15), respiration in darkness (10), the conclusion (5).
Lesson 1	50 min	As above, with written predictions at the start and the plant-free control running in parallel.
Lesson 2	50 min	The light-intensity series: maximum, shaded with one sheet, with two, darkness. The hunt for the compensation point — the intensity at which the line goes flat.
Lesson 3	50 min	Limiting factors (light, CO ₂ , temperature), the rate calculations, the rate-against-intensity graph, and the discussion of the limits of the forest analogy.

The plant decides whether the lesson works

Choose and try the plant before the lesson. A small plant with few leaves in a large jar gives a gradient so small it is lost in the noise; a large plant in a small jar gives a dramatic fall in five minutes.

The practical rule: **fill about a third of the jar's volume with leaves** — basil, mint, ivy or fresh spinach all work well — use a 2–5 litre jar and a bright light. **Do a ten-minute trial run** and check one thing: that the CO₂ really does *fall* in the light. If it does not, add leaves or move the lamp closer before the lesson, not during it.

And water the plant the day before. A thirsty plant closes its stomata, and a plant with closed stomata will not photosynthesise however much light you give it — which is, incidentally, an excellent explanation if it happens.

3. Materials

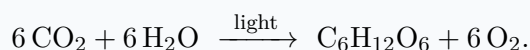
Item	Per group	Notes
Magic Box with SCD41 and the light sensor	1	Both sensors are needed. The box goes <i>inside</i> the jar, so check that it fits before the lesson.
Laptop with the app	1	WiFi is far more convenient: a cable coming out of the jar spoils the seal.
Jar or clear box, 2–5 L, with a lid	1–2	The second, without a plant, is the control — the most valuable thing you can add.
Fresh green plant	1	Basil, mint, ivy or spinach. About a third of the jar's volume in leaves. Watered the day before.
Bright light source	1	An LED desk lamp, or a sunlit window. LED is preferable : it does not heat the jar, and temperature is a limiting factor you do not want to change by accident.
Sheets of white paper	2–3	For shading in steps: one sheet, two, three. The simplest and most repeatable way to halve the light.
Black cloth or a cardboard box	1	For complete darkness, in the respiration experiment.
Tape or modelling clay	1	For sealing the lid around anything that leaves the jar.
Stopwatch	1	Rates are computed from gradients, and gradients need measured time.

Preparation. Choose and try the plant, as the tip above says. Assemble the jar with the box inside and check the seal: close it *empty*, with the box running, and watch for ten minutes — **if the CO₂ rises visibly in an empty, closed jar, you have a leak or a box drawing room air**, and it has to be fixed before the lesson. Set the light source at a fixed, marked distance and cut the shading sheets.

4. Scientific background for the teacher

The two equations, and why it matters that they are reverses

Photosynthesis:



Cellular respiration:



As overall equations they are exactly the reverse of each other — and that observation is where the whole activity comes from.

The plant does both, all the time. Cellular respiration never stops: it happens in every living cell, day and night. Photosynthesis happens only in the light, in cells with chloroplasts.

What the sensor measures is neither process but their difference, the *net* rate:

$$\text{net CO}_2 \text{ rate} = \text{respiration} - \text{photosynthesis}.$$

From this come the three regimes the class will see. In bright light, photosynthesis exceeds respiration and **the CO₂ falls**. In darkness, photosynthesis is zero, respiration remains, and

the CO₂ rises. And at some intermediate intensity the two are exactly equal, the net rate is zero and **the line goes flat:** this is the *light compensation point*.

The compensation point is a real quantity, tabulated in the literature for different species, and it is far lower than pupils expect — typically 10–50 $\mu\text{mol m}^{-2} \text{s}^{-1}$ of photosynthetically active radiation, which is of the order of a few hundred to a thousand lux for a shade plant. **That it can be found experimentally, in a school lesson, is the best thing in this activity.**

4.1 Why lux is not the same quantity as photosynthetically active radiation

This is the honest limit of the rig and it must be told to the class. Plants use **photosynthetically active radiation** (PAR), the number of photons between 400 and 700 nm, measured in $\mu\text{mol m}^{-2} \text{s}^{-1}$. The light sensor measures **illuminance**, in lux, which is radiant power weighted by the sensitivity of the human eye — peaking in the green, exactly where chlorophyll absorbs *least*.

The two can be converted into each other only through a factor that depends on the spectrum of the source: roughly 0.019 for daylight, but appreciably different for a white LED or an incandescent bulb. **Comparisons within one series, with the same lamp, are therefore valid; absolute values in lux cannot be compared with compensation points from the literature.** A good pupil should be able to state this.

4.2 Limiting factors, and how to recognise a plateau

At any moment the rate of photosynthesis is limited by *one* factor — the one in shortest supply. Increasing the others does not help. The three that matter here are:

- **Light.** At low intensities the rate rises almost linearly with illuminance. That is the stretch the intensity series measures.
- **CO₂.** In a closed jar the concentration *falls* as the plant works, so after a while **the CO₂ becomes the limiting factor itself** and the gradient eases off. It is a real effect, visible within a lesson, and it is why a gradient measured early must not be extrapolated.
- **Temperature.** The optimum is usually 25–35 °C. **An incandescent lamp heats the jar and changes this factor without anybody intending it** — one more reason for LED.

The sign that you have hit a limiting factor is **the plateau:** the rate stops rising although you raised the intensity. Pupils should hunt for the plateau, not avoid it.

4.3 The jar and the forest: the limits of the analogy

The app's final step links the activity to climate change, and the link is real — but it deserves to be placed correctly. A closed jar is not a forest: it has no living soil, no decomposers, no seasons and no exchange with the atmosphere. **What transfers is the mechanism, not the numbers.**

What can be said honestly: vegetation absorbs a significant share of annual human CO₂ emissions, and at night and in winter that same vegetation is a net source — exactly the regimes the class has just measured, at the scale of a jar. What *cannot* be said is that planting some number of trees offsets some quantity of emissions; that needs a budget the jar cannot support.

5. Lesson flow

The in-app wizard has **eight steps:** theory, the rig, three experiments, the quantitative analysis, limiting factors and the conclusion.

1. **Written predictions (5 min).** Before anything: *what will the CO₂ in the jar do in bright light? And in darkness? Is there a light level at which nothing happens?* The third question is the important one, and almost nobody answers “yes” before measuring.
2. **Theory (10 min).** The two equations on the board, one under the other, so that their being reverses is visible. Insist on **the net rate**: the sensor sees the difference, not the processes.
3. **Build the rig (10 min).** Plant and box in the jar, lid closed and sealed, lamp at the marked distance. **Ten minutes of settling**, during which the predictions for the intensity series are written.

Technical note — the four cards, and what exactly is read off the graph

The sensors. SCD41 for CO₂ (the CO₂ column) and VEML7700 for illuminance (the LUX column). The screen shows four cards: **CO₂** in ppm, **Light** in lx, **Temperature** in °C and **Humidity** in per cent. Temperature and humidity are not decorative: **they tell you whether the lamp is heating the jar**, that is, whether you have accidentally changed a limiting factor.

What is read — the gradient, not the value. The whole activity is done from *gradients*. The absolute CO₂ value in the jar depends on how much air was in it when you closed it and means nothing in itself. **The net rate is the CO₂ difference divided by the time**, in ppm/min, over stretches of at least five minutes.

How long you wait. **Ten minutes** after closing the jar, so that the air mixes and the sensor starts. **Five minutes after every change of light**, until the *gradient* settles — not the value, which by definition changes continuously.

The eight steps and their buttons. The green button changes its label at every step: **I understand the biochemistry!**, **Setup ready**, **Photosynthesis observed**, **Respiration observed**, **Intensities tested**, **Rates calculated**, **Limiting factors tested**, and finally **Finish**.

Exporting the data. At the measurement steps, **PNG** and **Excel** appear. **There is no CSV button inside the activity**; the CSV file is taken from the **Data & Export** page. **Export at the end of every lesson**, because the quantitative analysis is done on gradients computed from the file, not on readings taken off the screen.

4. **Experiment 1 — photosynthesis, in bright light (15 min).** Lamp at maximum. Record the CO₂ every five minutes and compute the gradient. **The CO₂ falls**: the plant is capturing carbon. Record the illuminance too, because it is the axis of the next series.
5. **Experiment 2 — respiration, in darkness (10 min).** The jar covered completely with the black cloth. **The CO₂ rises**: only respiration remains. The gradient here is *the respiration rate*, and it is used in every later calculation.
6. **Experiment 3 — the intensity series (lesson 2).** Maximum, one sheet, two sheets, three sheets, darkness. At every step record the illuminance and measure the gradient after five minutes of settling. **Look for the step at which the gradient is zero** — the compensation point.
7. **The quantitative analysis (lesson 3).** The graph of net rate against illuminance. The intercept with the horizontal axis is the compensation point; **the value at zero illuminance is the respiration rate** measured in experiment 2, which is a very satisfying internal check.
8. **Limiting factors and conclusion.** Discuss the plateau at high intensities and link it to the CO₂ falling in the jar. Close with the link — correctly placed — to the global carbon budget.

The investigation, in one table

The question. Is there a light intensity at which the plant neither consumes nor produces CO₂?

The hypothesis. Written **beforehand**: yes or no, and if yes, at roughly how many lux. Not changed afterwards.

What I change. One thing only: **the illuminance**, through sheets of paper added one at a time.

What I measure. **The gradient** of the CO₂ line, in ppm/min, over at least five minutes, **and** the illuminance inside the jar at every step.

What I keep the same. The same plant, the same jar, the same lamp distance, the same seal; **and the temperature** — which I check on the card at every step, because a lamp that heats changes a different factor from the one I am studying.

How many repeats. The series is run once in increasing order of shading and, if time allows, once in reverse. **If the compensation point comes out the same both ways**, the plant has not tired and the jar's CO₂ has not fallen far enough to become limiting itself.

What makes a reading valid — and what makes it an artefact. Valid is a gradient measured after five minutes of settling, with the temperature unchanged and the jar closed. An artefact is anything else: opening the jar between steps lets room air back in and erases everything; **a gradient measured over two minutes is noise**; and if the CO₂ has fallen well below the room value, the plant is already limited by CO₂ rather than light, and the series must be restarted with fresh air.

6. Analysis after the activity

6.1 What the data should look like

Condition	Typical net rate	What it means
Empty, closed jar (the control)	0 ± 2 ppm/min	The seal check. Any gradient here is subtracted from the others.
Bright light, healthy plant	−5 to −30 ppm/min	Photosynthesis clearly exceeds respiration.
Medium light (one sheet)	−2 to −10 ppm/min	Still net positive for the plant.
The compensation point	0 ppm/min	The line goes flat. Typically at a few hundred lux.
Complete darkness	+2 to +15 ppm/min	Respiration only. This is the respiration rate.
Bright light, after 30 min	the gradient eases off	The CO ₂ has itself become the limiting factor.
Temperature in the jar, LED lamp	+0 to +2 °C	Acceptable.
Temperature, filament lamp	+5 to +15 °C	You have changed two factors at once; the series is not clean.

6.2 From gradients to the compensation point, step by step

1. **Compute a gradient for every light step.** The CO₂ difference divided by the time, over at least five minutes. The sign matters: *minus* means the plant is consuming.

2. **Subtract the control's gradient**, if you ran a control jar. A leak of +1 ppm/min biases every value in the same direction.
3. **Plot net rate against illuminance**. Negative points below, positive points above, the horizontal axis passing through zero.
4. **Read the intercept**. The illuminance at which the curve crosses zero *is* the compensation point. Interpolate between the two neighbouring steps.
5. **Check internally**. The curve's value at zero illuminance should be approximately the respiration rate measured separately in darkness. **If the two do not agree, something changed between the experiments** — usually the temperature.

6.3 Questions to ask afterwards

1. The line was flat at a certain light level. Does that mean the plant was doing nothing? Explain what was actually happening in the leaves.
2. You measured illuminance in lux, but the plant uses photons between 400 and 700 nm. Why are the comparisons within your series still valid, and why can the absolute value not be compared with a textbook?
3. After thirty minutes in bright light the gradient eased off although the light had not changed. Which factor became limiting, and how would you check?
4. Why does the plant-free control matter? What exactly could you not rule out without it?
5. Your jar showed that vegetation absorbs CO₂ by day and releases it by night. What can you *not* conclude from that about the planet's forests?

6.4 Assessment ideas

- **Low stakes:** the pupil identifies the falling and the rising stretches on the graph and labels them correctly — photosynthesis dominant, respiration dominant.
- **Standard:** the pupil computes both rates, in ppm/min, and explains why the sensor measures a net rate rather than photosynthesis.
- **Extension:** the full rate-against-illuminance graph, with the compensation point found by interpolation, the internal check against the respiration rate, and a written critique of the lux-to-PAR conversion.

Common pitfalls and misconceptions

No control. Without an empty jar running in parallel, any CO₂ rise could be respiration *or* a leak. It is the cheapest possible control and it changes entirely what can be claimed.

Gradients measured over two minutes. The SCD41 gives a value every five seconds and has noise of its own; a two-minute gradient is noise with units. Five minutes minimum.

The filament lamp. It heats the jar by ten degrees and changes the temperature — another limiting factor — while you believed you were changing only the light. Use LED and **check the temperature card at every step**.

Opening the jar between steps. It lets room air back in and erases everything accumulated. If it must be opened, the series restarts.

Extrapolating a gradient measured early. After a while the jar's CO₂ falls far enough to become limiting itself and the gradient eases. The gradient at minute five does not predict minute forty.

“A flat line means nothing is happening.” It means the two processes cancel. It is precisely the misconception the activity demolishes.

“Plants only respire at night.” The commonest misconception at this level. Cellular respiration never stops; by day it is merely masked by photosynthesis.

A thirsty plant. A plant with closed stomata will not photosynthesise however much light it gets. Water it the day before.

Safety and classroom management

Lamps get hot, particularly filament ones, and here they stand beside a glass jar and beside shading paper. Keep the sheets away from the bulb, never cover the lamp, and let it cool before packing away. **Prefer LED** — for safety and for the cleanliness of the experiment alike.

Glassware. A large jar with an electronic box inside is heavy and slippery. It is held with two hands, stands on a stable surface away from the edge of the bench, and is checked for chips first.

Electronics in the jar. The box is not waterproof: do not put water in the jar and do not water the plant while the box is inside. If you use a potted plant, the pot must be dry on the outside.

Plants. Do not use toxic or irritant plants — common ivy is acceptable if handled with care, but spinach, basil and mint are better choices. Nothing is tasted. Washing hands after handling soil is the usual laboratory rule.

Cables. On USB the cable leaves the jar and spoils the seal; WiFi solves both the sealing problem and the trip hazard.

Adaptations for pupils with special educational needs

The graph read rather than computed. This is the main adaptation: the pupil does not compute gradients but *describes* the graph — where it falls, where it rises, where it is flat — and puts the right label on each stretch. **Recognising the three regimes is the objective**, and it is met with no arithmetic at all.

Axes drawn in advance. For the rate-against-illuminance graph, the sheet arrives with the axes drawn, scaled, and the zero line thickened. The pupil places four points and reads the intercept.

One gradient, worked in full. The first rate comes printed on the sheet, with every step, as a pattern. The rest are not pre-computed.

Three real roles for pupils with motor difficulties. The rig involves handling a heavy jar, but the roles that matter are done from a desk:

- **the temperature keeper** — checks the temperature card at every step and has the right to stop the series if the lamp is heating the jar; this is the role with the greatest scientific value, because they defend the “only one factor changed” condition;
- **the timekeeper** — announces the five minutes of settling and the start and end of each gradient window;
- **the scribe** — writes the values down and computes the gradients as they arrive.

Light sensitivity. The bright lamp can be unpleasant. It can be shielded at the sides with card so that it lights only the jar, and the roles above are done from the far side of the bench.

If things do not go to plan

The CO₂ does not fall in the light. Check, in order: are there enough leaves (about a third of the volume)? is the lamp close enough? was the plant watered? is the jar actually closed? **A thirsty plant or too few leaves are the usual causes**, and both are fixed in five minutes.

The CO₂ rises in the light as well as in the dark. You have a leak, or the box is

drawing room air. Run the jar *empty* for ten minutes: if it rises there too, the problem is the seal, not the plant.

You have no second jar for a control. Run the control *first*, with the same jar empty, for ten minutes, and use that gradient as a correction. It is less good than simultaneous, but far better than nothing.

You cannot find the compensation point. Your steps are probably too far apart. Add sheets one at a time and, if it still will not appear, move the lamp away instead of shading — varying the distance continuously gives finer resolution than sheets do.

You have no time for three lessons. Run the 50-minute route: photosynthesis and respiration are enough to demolish the central misconception. The compensation point is an extension, not the core.

The box will not connect. Do not stop the lesson. Use the “What the data should look like” table as a printed dataset: pupils compute both rates, plot the rate-against-illuminance graph from the given values and find the compensation point by interpolation. **The whole analysis — which is the lesson — is done without the equipment.** Measure again another day, after checking the cable, the port selected in the app and, if you work over WiFi, the network.

7. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are base addresses, not deep links, so they do not rot.

- **Royal Society of Biology** — <https://www.rsb.org.uk> — search for photosynthesis and limiting factors; material written for teachers, with tested laboratory variants.
- **PhET Interactive Simulations** — <https://phet.colorado.edu> — search for the photosynthesis and gases simulations; the first lets pupils vary light and CO₂ before they measure them.
- **Nuffield Foundation** — <https://www.nuffieldfoundation.org> — classic practical work on photosynthesis, useful for the sensor-free variants that run alongside.
- **NASA Climate** — <https://climate.nasa.gov> — search for the carbon cycle and the role of vegetation, for the final step, placed correctly.
- **Global Carbon Project** — <https://www.globalcarbonproject.org> — the global carbon budget, if a pupil wants the real figures behind the climate link.
- **Institute of Physics** — <https://www.iop.org> — search for material on photometry and radiometry, for the difference between lux and photosynthetically active radiation.

8. Assessment

The exit ticket, identical to the one on the pupil sheet: *what is happening in the leaves when the CO₂ line is flat?* Collected in the last two minutes, it says more about the class’s understanding than any mark.

For a more formal judgement, use the **common three-axis rubric** — *measurement practice*, *data interpretation* and *scientific reasoning*. At Scientist level the three axes look like this:

- **Measurement practice** — the pupil runs a control, respects the five minutes of settling at every step, checks the temperature so as not to change two factors at once, and does not open the jar between steps.
- **Data interpretation** — the pupil works with *gradients* rather than absolute values, subtracts the control’s gradient, and reads the compensation point by interpolation rather than by

guessing.

- **Scientific reasoning** — the pupil explains why the sensor measures a net rate, recognises the plateau as the sign of a limiting factor, and states the limit of the lux-to-PAR conversion.

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

9. Pupil sheet

The sheet the pupils fill in is a separate document: `docs/en/student_sheets/co2_and_photosynthesis.pdf`. It has two pages — the predictions, the rig and the two basic experiments on the first; the intensity series, the rate-against-illuminance graph and the conclusion on the second — so it is printed double-sided, one sheet per pupil. It is built on the three layers — what I see on the screen, what I do on my own, the conclusion I write.

Three things to know before printing it. **The sheet asks for gradients, not values**, with columns for the CO₂ at the start and end of the window and for its duration: the absolute value in the jar means nothing, and a sheet that asked for it would teach the wrong thing. **The control row is printed first**, even though some classes will not fill it in — its presence on the paper is what makes the question “how do you know it is not a leak?” arise by itself. **The rate-against-illuminance graph comes with its axes drawn and its zero line thickened**, because the compensation point is *the intercept with that line*, and if the line is not visible the intercept is not looked for. The tables are not repeated here precisely so that the two documents cannot drift apart: there is one printable version, and it is the sheet.